



Innovative temperature-responsive membrane with an elastic interface for biofouling mitigation in industrial circulating cooling water treatment

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ABSTRACT

To address the issues of scaling caused by heat and water evaporation in regard to circulating cooling water (CCW), TFC membrane filtration systems have been increasingly considered for terminal treatment processes because of their excellent separation performance. However, membrane biofouling phenomenon significantly hinders the widespread utilization of TFC membranes. In this study, to harness the thermal phenomenon of CCW and establish a stable and durable multifunctional antibiofouling layer, temperature-responsive PnIPam and the spectral antibacterial agent Ag were organically incorporated into commercially available TFC membranes. Biological experimental findings demonstrated that above the lower critical solution temperature (LCST), the contraction of PnIPam molecular chains facilitated the inactivation of bacteria by the antibacterial agent, resulting in an impressive sterilization efficiency of up to 99 %. XDLVO analysis revealed that below the LCST, the establishment of a hydration layer on the functional interface resulted in the creation of elevated energy barriers, effectively impeding bacterial adhesion to the membrane surface. Consequently, a high bacterial release rate of 98.4 % was achieved on the low-temperature surface. The alterations in the functional membrane surface conformation induced by temperature variations further amplified the separation between the pollutants and the membrane, creating an enhanced "elastic interface." This efficient and straightforward cleaning procedure mitigated the formation of irreversible fouling without compromising the integrity of the membrane surface. This study presents a deliberately engineered thermoresponsive antibiofouling membrane interface to address the issue of membrane fouling in membrane-based CCW treatment systems while shedding new light on the mechanisms of "inactivation" and "defense."

1. Introduction

In the realm of industrial water usage, circulating cooling water (CCW) has emerged as the highest water-consuming category (Wagner et al., 2018; Zhu et al., 2021). According to the statistics, a staggering 14, 800 m³/year of water is wasted by a 500 t cooling tower, thereby exacerbating the predicament of water scarcity (Meese et al., 2022). Owing to the thermal presence and water evaporation within the system, it becomes highly susceptible to the proliferation of biological organisms and the formation of deposits (Wagner et al., 2022). In conventional treatment approaches, chemical reagents are typically added to control pollution; however, this not only increases the treatment cost but also has the potential to cause secondary pollution (Wang et al., 2019a). Additionally, a substantial amount of fresh water is required to sustain the system's operation (Luo et al., 2024). Hence, the development of

CCW reuse technologies that can effectively curtail the discharge of wastewaters is imperative.

Membrane filtration systems have unique structural and physico-chemical properties and show immense potential in the domain of water treatment. These systems are widely recognized as an optimal solution to address the pressing issue of water scarcity (Liu et al., 2017). Compared to UF membranes, TFC membranes exhibit higher separation efficiency, and the produced water can be partially reused as freshwater in the CCW system to maintain system stability (Guo et al., 2022; Lee et al., 2016a). However, membrane biofouling has a significant and irreversible impact on the water production capacity of these membranes, resulting in increased energy consumption and high cleaning costs, thereby impeding the progress of membrane technology (Tong et al., 2020). Therefore, effective control of TFC membrane biofouling has become a critical imperative.

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